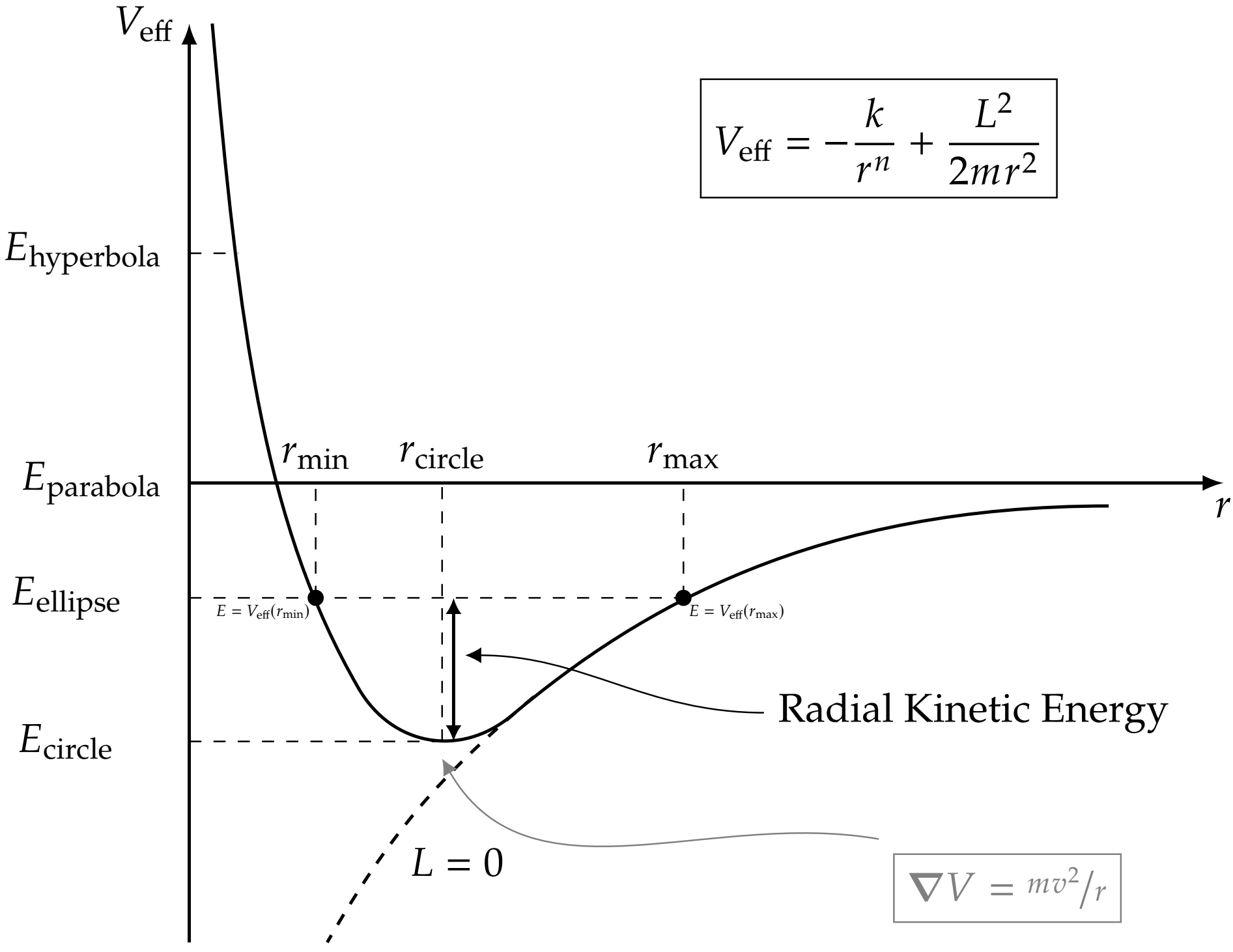

V_{eff}
$V_{\text{eff}} = -\frac{k}{r^n} + \frac{L^2}{2mr^2}$
$E_{\text{hyperbola}}$
E_{parabola}
E_{ellipse}
E_{circle}
r_{min}
r_{circle}
r_{max}
r
$E = V_{\text{eff}}(r_{\text{min}})$
$E = V_{\text{eff}}(r_{\text{max}})$
Radial Kinetic Energy
$L = 0$
$\nabla V = mv^2/r$